

Heaven's Light is Our Guide

Rajshahi University of Engineering and Technology

Department of Electrical & Electronic Engineering



Assignment

Course Code: EEE 2211

Course Title: Measurement and Instrumentation

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Question: What is the operating principle of Hall-effect transducers? Enlist some of their applications.

Answer: If a strip of conducting material carries current in the presence of a transverse magnetic field as shown in Fig. an emf is produced between two edges of conductor. The magnitude of the voltage depends upon the current, flux density and a property of conductor called "Hall Effect Co-efficient".

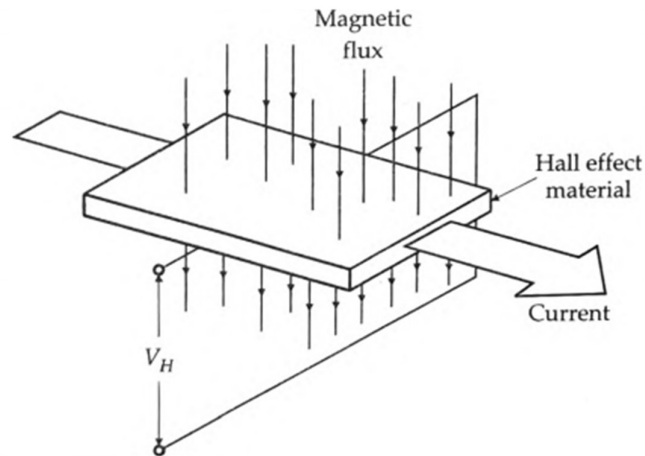


Fig: Hall Effect material.

The emf may be measured after amplification. Hall effect elements are extensively used in magnetic measurements. They can also be used for sensing current.

When a conductive plate is connected to a circuit with a battery, then a current starts flowing. The charge carriers will follow a linear path from one end of the plate to the other end. The motion of charge carriers results in the production of magnetic fields. When a magnet is placed near the plate, the magnetic field of the charge carriers is distorted. This upsets the straight flow of the charge carriers. The force that upsets the direction of the flow of charge carriers is known as the Lorentz force.

Due to the distortion in the magnetic field of the charge carriers, the negatively charged electrons will be deflected to one side of the plate and positively charged holes to the other side. A potential difference, known as the Hall voltage will be generated between both sides of the plate which can be measured using a meter.

The Hall voltage represented as V_H is given by the formula:

$$V_H = \frac{IB}{qnd}$$

Here,

I is the current flowing through the sensor

B is the magnetic field strength

q is the charge

n is the number of charge carriers per unit volume

d is the thickness of the sensor.

Applications of Hall Effect Transducer

1. **Magnetic to Electric Transducer:** The Hall Effect Transducer can be used as a Magnetic to Electric transducer. A semiconductor plate is inserted into the magnetic field to be measured.
2. **Measurement of Displacement:** The Hall effect element can be used for the measurement of the location or displacement of a structural element i.e., it can serve as an indirect acting position displacement or proximity transducer in cases where a change of geometry of a magnetic structure causes a change of magnetic field strength.
3. **Measurement of Current:** The device serves to measure current in a conductor without the need for interrupting the circuit and without making an electrical connection between the conductor circuit and the meter.
4. **Measurement of Power:** Power measurements are made in high voltage circuits connecting the wattmeter to the circuit through current and potential transformers